Automatic Circuit Discovery for Explainable Network Intrusion Detection

Adapting ACDC for Mechanistic Interpretability in NIDS

Based on: Arthur Conmy, Augustine N. Mavor-Parker, Aengus Lynch, Stefan Heimersheim, Adrià Garriga-Alonso

Presented by: **Anbuselvi S**

Presentation Outline

- Motivation: Explainability in NIDS
- Mechanistic Interpretability Concepts
- Automatic Circuit Discovery (ACDC)
- Adapting ACDC to NIDS
- System Design
- Expected Contributions
- Limitations and Future Work

Motivation: Explainability in NIDS

- Deep learning based NIDS achieve high detection accuracy
- However, modern NIDS models behave as **black boxes**
- Existing explainability methods (SHAP, LIME):
  - provide feature attribution
  - lack causal guarantees
  - fail to explain *internal computation*
- Security-critical systems require:
  - trustworthy decisions
  - auditability
  - forensic interpretability

Behavior Definition in NIDS

- In mechanistic interpretability, a **behavior** is:
  - a well-defined input–output pattern
  - measurable via a task-specific metric
- In NIDS, behaviors correspond to:
  - DDoS detection
  - Port scanning detection
  - Brute-force or probe attacks
- Each behavior is isolated using:
  - attack-specific datasets
  - prediction confidence or logit difference

What Is a Circuit?

- A **circuit** is a sparse subgraph of the model
- It captures the internal computation responsible for a behavior
- In transformers: attention heads and MLP blocks
- In NIDS models:
  - feature interaction paths
  - temporal dependencies
  - hidden neuron activations
- Goal: isolate the minimal computation needed for attack detection

Scalability Challenges in Circuit Discovery

- Traditional circuit discovery relies on:
  - manual inspection
  - human intuition
  - trial-and-error patching
- This approach does not scale to:
  - deep NIDS architectures
  - multiple attack classes
  - large feature spaces
- **ACDC automates this step**

Automatic Circuit Discovery (ACDC)

- ACDC is a post-hoc, causal interpretability method
- Operates on a **trained and frozen model**
- Key idea:
  - represent the model as a computational graph
  - iteratively remove causally irrelevant connections
- Output: minimal, behavior-preserving circuit

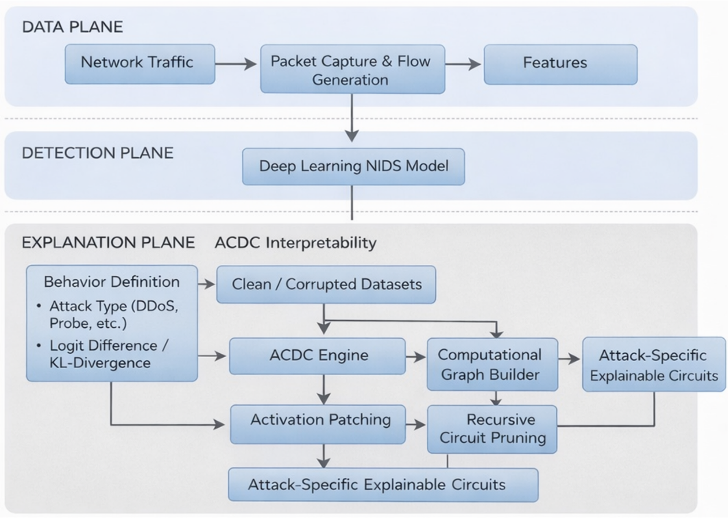
ACDC Pipeline for NIDS

- Step 1: Define attack-specific behavior
- Step 2: Construct computational graph of NIDS
- Step 3: Generate clean and corrupted traffic
- Step 4: Apply activation patching
- Step 5: Recursively prune unimportant edges
- Step 6: Extract attack-specific circuit

Activation Patching

- Activation patching tests causal importance
- Replace internal activations with corrupted versions
- Measure impact on attack prediction
- If prediction unchanged:
  - component is not causally necessary
- Forms the basis of edge removal in ACDC

System Design: ACDC-Enabled NIDS



System architecture integrating deep learning NIDS with ACDC-based circuit discovery

Expected Research Contributions

- Mechanistic interpretability for NIDS
- Attack-specific circuit extraction
- Causal explanations beyond feature attribution
- Improved trust and forensic analysis
- Foundation for robust and debuggable NIDS

Limitations and Future Work

- ACDC is computationally expensive
- Requires careful choice of corrupted data
- Current focus on offline analysis
- Future work:
  - scaling to real-time systems
  - multi-attack joint circuits
  - robustness against adaptive attacks

Thank You!